



Fig. 3: Test set-up for CMTI

requirements of the IPM driver are governed by the switching speed required for the IPM module in question, or the pulse-width modulation (PWM) dynamic range. For example, with a 20kHz switching period (typical for IGBT-based IPMs) and a design requiring modulation of the switching pulse from 10 per cent to 99 per cent, the minimum pulse width required will be approximately 5 µs. Correspondingly, if the minimum pulse width required is 1 per cent, a minimum pulse of 0.5 µs would be needed. The pertinent calculation is:

$$t = (1/\text{Switching frequency}) \times \text{Minimum pulse width required}$$

A good rule of thumb is that the maximum propagation delay should be less than the minimum pulse width required.

The next switching issue of concern is dead time. Most switching applications arrange their output switches in a 'push-pull' configuration. In this type of switch arrangement, it is extremely important to assure that the low-side switch is turned off before the high-side switch is turned on, and vice versa. If both switches are turned on at the same time, a short is created across the high-voltage power supply. At best, this results in tremendous stress on the output power components, if confined to a very short time. At worst, the result is smoke and fire.

In order to minimise this momentary high-current flow, commonly known as 'shoot-through', the turning on of one switch is delayed from the turn-off of the complementary switch; the delay between these two switching

events is known as dead time.

When both the switches are off, nothing happens. As a result, increasing dead time decreases the efficiency of motor drives and solar inverters. Therefore it is desirable to have as much dead time as needed to prevent shoot-through, but no more.

The optimal case for dead time would be a value of zero, where one switch closes instantaneously and the other switches on instantaneously. This type of behaviour is possible to achieve with simulations, but not with real components.

In terms of the IPM driver, ignoring any delays in the switches themselves, a zero delay would be required when the high-to-low propagation delay (tphl) is exactly equal to the low-to-high propagation delay (tph). Again, this is not a realistic condition as there is some variance between tphl and tph.

Consequently, it is necessary to first calculate the maximum $\Delta t_{ph} = t_{phl}(\text{max}) - t_{phl}(\text{min})$, and minimum $\Delta t_{ph} = t_{phl}(\text{min}) - t_{phl}(\text{max})$. Given this, the maximum dead time that can be accounted for by the IPM driver is the difference between maximum Δt_{ph} and minimum Δt_{ph} (max. dead time = maximum $\Delta t_{ph} - \text{minimum } \Delta t_{ph}$). In short, most current IPM driver applications, which are limited to switching frequencies of less than 20 kHz, can be satisfied with IPM drivers with data rates of 1 Mbps/1 MBd or 5 Mbps/5 MBd.

Another important consideration when specifying IPM drivers is their ability to provide isolation from hazardous electrical circuits, as well as

noise isolation between noisy circuits, such as IPM modules, and sensitive control circuits.

Noise isolation is not as easy to quantify as safety isolation. There are two parameters that can be used

to quantify the value of an IPM driver with regards to this most important function. One is the coupling capacitance from the primary side of the IPM driver to the secondary side. The larger the coupling capacitance, the less effective the IPM will be in terms of noise isolation; however, capacitance in most optocouplers is given as <<1 pF. Such measurements are difficult to make and appreciate.

Arguably, a more meaningful parameter that quantifies the noise isolation capability of a device is common-mode transient immunity (CMTI). The test set-up for this parameter is shown in Fig. 3.

The test methodology is conceptually simple. A high transient voltage potential is placed between the primary and secondary sides, and the amount of the transient that leaks across the isolation barrier measured. An IPM module should have a high CMTI greater than or equal to 10 kV/µs.

There are other parameters of concern to designers when designing opto-isolators, such as operating temperature and isolation voltage values; however, the issues of timing and noise isolation are of paramount importance to IPM drivers in particular. A careful application of opto-isolated IPM drivers can greatly increase reliability and reduce overall costs in new high-power switch-mode designs. ●

The author is senior product marketing manager, optocouplers, at Vishay Intertechnology. He holds a bachelor's degree in electrical engineering and a master's degree in economics from San Jose State University, USA